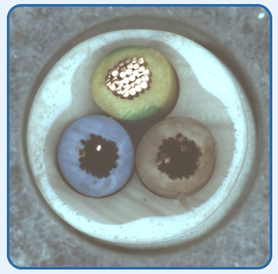
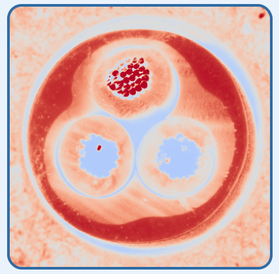
① Input

Input image x



raw industrial image

ImageNet normalisation



mean/std standardised

normalised

input image

② Backbone (Frozen)

DINOv2 ViT-B/14

frozen backbone · no gradient
multi-scale patch features (P3, P4)

P3

resize → 392 px
patch stride 14
→ 28 × 28 tokens
C = 768 channels

P4

resize → 224 px
patch stride 14
→ 16 × 16 tokens
C = 768 channels

Fixed random 1×1 projection

d = 256 · one frozen basis, never trained

$\varphi_{P3}, \varphi_{P4}$

③ Normal Reference

Position-wise Gaussian reference

$\mu_k = E[\varphi(x)_k]$
per-position normal mean

$\Sigma_k = \mathrm{diag}(\sigma^2_k)$
shrinkage λ = 0.02

$\mathrm{ref_smooth}(\mu_k)$
spatial smoothing (off)

normal (defect-free) images only
no labels, no gradients used

μ_k, Σ_k

④ Scoring & Multi-scale Fusion

Diagonal Mahalanobis distance

$$d_k = (\varphi_k - \mu_k)^T \Sigma_k^{-1} (\varphi_k - \mu_k)$$

per-position standardisation
sigmoid → score in (0, 1)
diag Σ_k from the normal reference

d_k · per scale

Multi-scale mean fusion

$$\hat{z} = (\mathrm{Up}(\hat{z}_{P3}) + \mathrm{Up}(\hat{z}_{P4})) / 2$$

resample → 28×28 · equal weights 1/2 + 1/2

$\hat{z}$

⑤ PSC

Texture energy T

Haar high-frequency energy
of the original input image

T

$$\alpha = \mathrm{clip}(a_0 + a_1 \cdot T, 0, 1)$$

$a_0 = 0.6$, $a_1 = 0$ → gate OFF

α

$$\hat{z}_{cal} = (1 - \alpha) \cdot \hat{z} + \alpha \cdot \mathrm{Blur}_K(\hat{z})$$

K = 33 · Gaussian low-pass (DEPLOYED)
α = a0 = 0.6; texture gate off (a1 = 0)
fixed low-pass smoothing

$\hat{z}_{cal}$

⑥ Output

Ours: anomaly map ẑ_cal

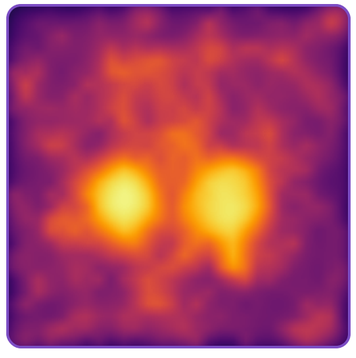


Image-level score

top-q mean (q = 0.5%)
dense localisation